

BOKARO STEEL PLANT — ILLUSTRATIVE SAMPLE

Standard Operating Procedure

SOP No: BSP-HSM-021

Title: Work Roll Change in Finishing Mill Stands

Department: Hot Strip Mill

Revision: 02

Date: 12-Apr-2024

1. PURPOSE

To ensure safe and quick changing of work rolls in finishing stands F1 to F7 of the Hot Strip Mill.

2. SCOPE

Applies to roll shop crew, mill operators, electrical and mechanical maintenance staff.

3. PREREQUISITES

- Mill stopped and "Roll Change Mode" selected on the pulpit HMI
- Electrical isolation of main drive and screw-down as per LOTO SOP BSP-EL-005
- Descaling water pumps isolated and pressure at 0 bar confirmed on gauge
- New roll set pre-checked in the roll shop, chocks fitted, roll diameter recorded

4. PPE

- Safety helmet, safety goggles, cut-resistant gloves, safety boots
- Ear protection (noise in the stand area exceeds 90 dB)

5. ROLL CHANGE PROCEDURE

Step 1 — Open the stand to maximum gap and lower the work-roll balance.

Step 2 — Connect the roll change car; confirm the latch indicator is green.

Step 3 — Pull the old roll set out; keep a distance of at least 3 metres from the moving car.

Step 4 — Push the new roll set in and engage the chock locks.

Step 5 — Restore hydraulics and run a zero-level calibration before rolling.

6. HAZARDS

- Crushing between chocks and the stand housing
- Hot surfaces: rolls may be above 60°C for 2 hours after rolling stops
- High-pressure water (descaling header up to 180 bar)

7. HANDOVER

The roll change supervisor removes personal locks only after all crew members confirm they are clear of the stand.